

ShopEasy Consumer Intelligence Funnel & Sentiment Analysis

Tools: SQL Server · Python · Power BI · NLP

1. Introduction

This project is an interactive Power BI dashboard built on a simulated retail dataset for ShopEasy. It analyses customer journey behaviour, marketing engagement, and sentiment data to identify conversion bottlenecks, product health risks, and declining engagement trends. The dashboard highlights KPIs such as conversion rates, sentiment distribution, and product health scores, while providing funnel analysis, product performance insights, and actionable recommendations.

2. Design Components

Data Sources

- Customer journey dataset (Views, Clicks, Purchases, Drop-offs).
- Customer reviews (text + star ratings).
- Marketing engagement data (views, clicks, likes, channel mix).

Data Model

- Relationships between Journey, Reviews, and Engagement tables.
- Centralized Date dimension for cross-filtering across fact tables.

Key Measures (DAX / SQL):

- Conversion Rate = $\text{Purchases} \div \text{Clicks}$.
- Sentiment % = $\text{Positive} \div \text{Total Reviews}$.
- Product Health Score = Composite of Conversion, Rating, Sentiment.
- Funnel Drop-off = $\text{SUM}(\text{Drop-offs at Checkout})$.

3. Dashboard Features

- **Top KPIs:** Overall conversion rate (9.6%), sentiment distribution (61.6% positive, 31% negative), average rating (3.7/5).
- **Conversion Funnel:** View → Click (51%), Click → Purchase (19%), Drop-off concentration at Checkout (96%).
- **Marketing Performance:** 12M views, 2M clicks, 529K likes; CTR ~20% across all channels.
- **Customer Sentiment:** Complaint categories (Unmet Expectations 31%, Cost 21%, Performance 18%, Delivery 17%).

4. Program Flow

1. Load raw journey, review, and engagement data into SQL Server.
2. Validate and clean data (duplicates, casing, whitespace, column splits).
3. Run NLP sentiment pipeline (VADER + keyword extraction).
4. Create calculated measures and composite scoring in Power BI.
5. Build visuals (funnel charts, sentiment breakdown, product scorecards).
6. Apply slicers for dynamic filtering (date, product, channel).
7. Generate insights and recommendations.

5. Example Output

Conversion Funnel:

- View → Click: 51%
- Click → Purchase: 19%
- Drop-offs at Checkout: 96%

6. Key Takeaways

- **Conversion Bottleneck:** Strong top-funnel engagement, but sharp collapse at Checkout.
- **Sentiment Gap:** 31% negative reviews, driven by unmet expectations in product descriptions.
- **Product Risk:** Running Shoes flagged At Risk despite highest click volume.
- **Marketing Decay:** Engagement peaks in Q1 but declines uniformly by Q4.

7. Real-World Applications

- **Sales Strategy:** Simplify checkout UX to improve completion rates.
- **Marketing:** Pause promotions on At Risk products until issues resolved.
- **Product Management:** Rewrite product descriptions to reduce unmet expectations.
- **Customer Experience:** Leverage sentiment analysis to address dissatisfaction drivers.

8. Learning Source

This project was inspired by combining SQL data validation, Python NLP sentiment analysis, and Power BI dashboard design. It extends concepts from data cleaning, text analytics, and business intelligence tutorials into a unified consumer intelligence framework.